

Module: MATH97113
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EXAMINATION FOR THE MSc IN MATHEMATICS AND FINANCE

Thursday 13 January 2022

MATH97113 Stochastic Processes

Date: Thursday 13 January 2022

Time: 10am - 1pm

Time allowed: 3 Hours

This paper has 4 *Questions*.

Candidates should start their solutions to each question in a new main answer book.

Supplementary books may only be used after the relevant main book(s) are full.

- Calculators may not be used.
- Credit will be given for all questions attempted.
- Question 1: 25 points
- Question 2: 25 points
- Question 3: 25 points
- Question 4: 25 points
- The rigour and clarity of your answers will be taken into account in the final grade.
- Each problem is independent of the others.
- Red or green ink or correction fluid (such as Tipp-Ex) may not be used.

DO NOT OPEN THIS PAPER BEFORE THE EXAM STARTS.

1. Measure Theory

Let $(\Omega, \mathcal{F}, \mu)$ a measure space.

- (a) Let f be a simple function that admits the following standard representations:

$$f(x) = \sum_{j=1}^M c_j 1_{A_j}(x), \quad f(x) = \sum_{i=1}^N d_i 1_{B_i}(x),$$

for some constants $\{c_j\}_{j=1}^M$, $\{d_i\}_{i=1}^N$ and sets $\{A_j\}_{j=1}^M$, $\{B_i\}_{i=1}^N$ in $\mathcal{F}$. Prove that $\int f d\mu$ is the same under both representations. (10 marks)

- (b) Let $\{f_n\}_{n \geq 0}$ be a sequence of measurable functions in $L^1(\Omega, \mathcal{F}, \mu)$ that converge a.e. to a measurable function f , which is also in $L^1(\Omega, \mathcal{F}, \mu)$, as $n \rightarrow \infty$. Prove or disprove by presenting a counterexample the following:

$$\lim_{n \rightarrow \infty} \int f_n d\mu = \int f d\mu.$$

(5 marks)

- (c) Under the same setting as in part (b) suppose now that $|f_n| \leq g$ for any $n \geq 1$, where g is in $L^p(\Omega, \mathcal{F}, \mu)$ for some $p > 1$. Prove or disprove by presenting a counterexample that

$$\|f_n - f\|_{L^p(\Omega, \mathcal{F}, \mu)} = 0, \quad \text{as } n \rightarrow \infty.$$

(10 marks)

(Total: 25 marks)

2. Conditional Expectation

Let X and $\tilde{X}$ be two random variables on the probability space $(\Omega, \mathcal{F}, P)$. Let $\mathcal{G} \subset \mathcal{F}$, be subsigma-field.

- (a) Assume that $X \leq \tilde{X}$ a.s. Prove that

$$E[X|\mathcal{G}] \leq E[\tilde{X}|\mathcal{G}], \quad \text{a.s.}$$

(5 marks)

- (b) Under the same setting as in (a) let Z be bounded and $\mathcal{G}$ -measurable random variable. Prove that

$$E[XZ|\mathcal{G}] = ZE[X|\mathcal{G}], \quad \text{a.s.}$$

(10 marks)

- (c) Denote by $L_0^2(\mathcal{F})$ the subclass random variables Y in $L^2(\Omega, \mathcal{F}, P)$ with $E(Y) = 0$. Let $\mathcal{G} \subset \mathcal{F}$, be subsigma-field. For a chosen X from $L_0^2(\mathcal{F})$ solve the following optimisation problem

$$\inf_{Y \in L_0^2(\mathcal{G})} \text{Var}(X - Y).$$

(10 marks)

(Total: 25 marks)

3. Martingales

Let X_i^n , $n, i \geq 1$, be i.i.d **nonnegative integer-valued** random variables. Define the process $Z = \{Z_n\}_{n \geq 0}$ with $Z_0 = 1$ and

$$Z_{n+1} = \begin{cases} X_1^{n+1} + \dots + X_{Z_n}^{n+1}, & \text{if } Z_n > 0, \\ 0 & \text{if } Z_n = 0. \end{cases}$$

The process Z is known as a branching process. We think of Z_n as the number of individuals in the population at generation n . Each individual i at generation n gives birth to X_i^{n+1} offsprings at generation $n+1$. The common distribution of the i.i.d. random variables $\{X_i^k\}_{i \geq 1, k \geq 1}$ is called the offsprings distribution. For any $\xi > 0$, we denote its probability generating function by:

$$G(\xi) = E[\xi^{X_i^k}].$$

- (a) Assume that there exists $\xi > 0$ such that $G(\xi) = \xi$. Prove that $\{\xi^{Z_n}\}_{n \geq 1}$ is a martingale with respect to the natural filtration. (10 marks)
- (b) For any ξ as in (a), prove that there exists a random variable M_∞ such that

$$M_\infty = \lim_{n \rightarrow \infty} \xi^{Z_n}, \quad \text{a.s.}$$

(5 marks)

- (c) Show that $E[M_\infty] \leq \xi$ for any ξ as in (a) and that for $\xi \leq 1$

$$E[M_\infty] = \xi.$$

(10 marks)

(Total: 25 marks)

4. Stochastic Calculus

Let $B = (B_t)_{t \geq 0}$ be a one-dimensional standard Brownian motion.

- (a) Use the fact $E[Z^4] = 3\sigma^4$ for a mean-zero Gaussian random variable Z with standard deviation σ and Itô's Formula in order to calculate

$$E \left[\int_0^t B_s^2 ds \right], \quad \text{for all } t \geq 0. \quad (1)$$

Use also Fubini's Theorem in order to compute (1) directly and compare your answers.

(10 marks)

- (b) Suppose that $g : \mathbb{R} \rightarrow [0, \infty)$ is a bounded continuous function and define

$$R_t = \int_0^t g(s) dB_s.$$

Prove that

$$M_t = R_t^2 - \int_0^t g(s)^2 ds, \quad t \geq 0,$$

is a martingale.

(15 marks)

(Total: 25 marks)